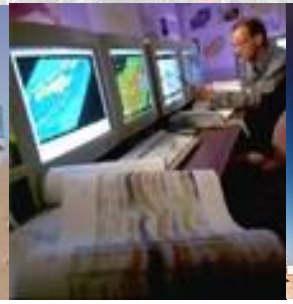
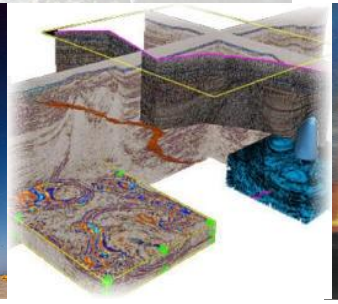
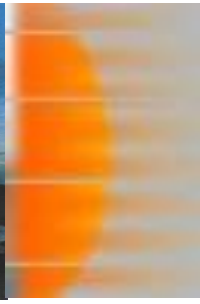
ENG-3020
RENEWABLE ENERGY SOURCES









Introduction,
Basic concepts, definitions
and
energy units

What is Energy?

- ***Energy is the ability to do work***

or

Power derived from the chemical or physical sources to provide light, heat or to work machines

Energy comes in different forms:

- Heat (thermal)
- Light (radiant)
- Motion (kinetic)
- Electrical
- Chemical
- Nuclear energy

What is Energy?



$E = mc^2$

- E stands for "energy,"
- M denotes "mass"
- C is the speed of light.

- What is energy?
- What is electricity
- Can electricity be stored?
- Is electricity material or a product?
- How can electricity be produced?

Energy: cannot be created or destroyed but can be converted into different forms.



What is Energy?

- The universe is made up of matter and energy.

- In physics, energy is the ability to do work, or the ability to move or the ability to change the objects. In fact, the amount of energy something has refers to its capacity to cause things to happen.

- Energy has a few important properties. First, **energy is always conserved, it cannot be created or destroyed**. However, it can be transferred between objects or systems by the interactions of forces, and it can be converted from one form to another.

- The two most common or basic forms of energy are kinetic energy and potential energy.

Why do we need Energy?

- Energy can be found everywhere.
- Energy demands are directly associated to the economic growth of a country.
- Energy growth is an indicator to well-being and prosperity of a country.
- Meeting the energy demand in a safe, sustainable and environmentally responsible manner is the key challenge for the modern world.
- There are approximately 8 billion people on earth who use energy in their daily life to make their lives easier, richer, productive, safer and healthier.

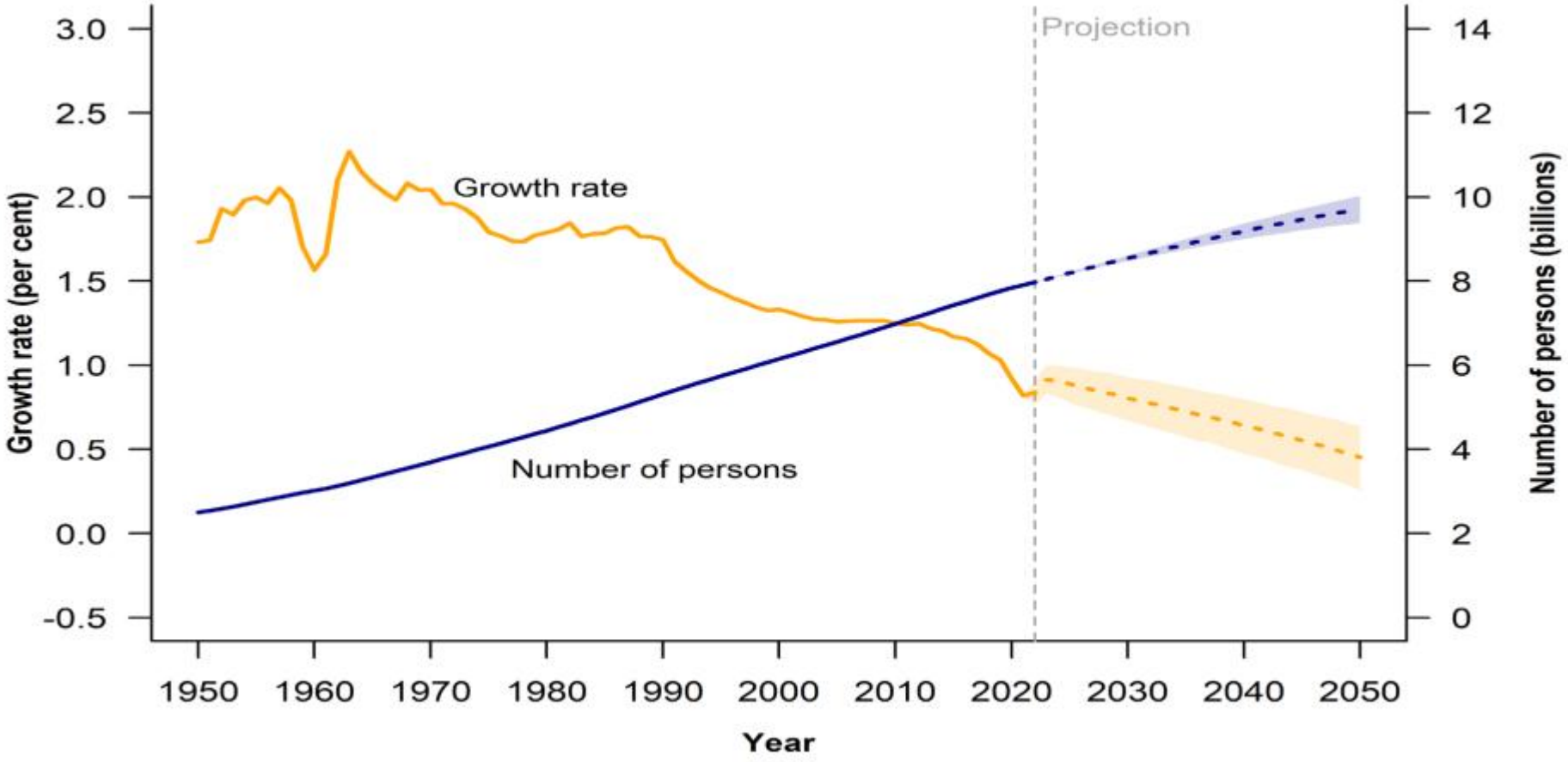
Why do we need Energy?

- By 2035 world's population is expected to be 8.5 billion where extra 0.6 billion people will need energy

- Energy consumption will increase accordingly

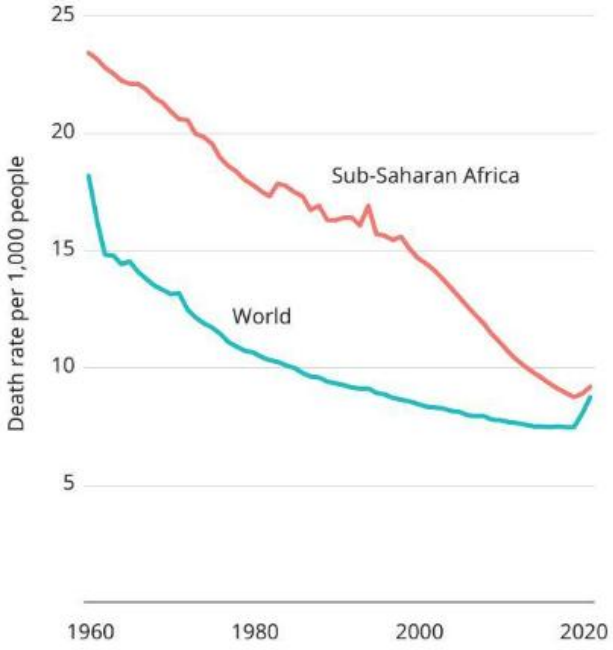
- Energy is the lifeblood of the global economy; crucial input to nearly all of the goods and services of the modern world.

World population and growth rate

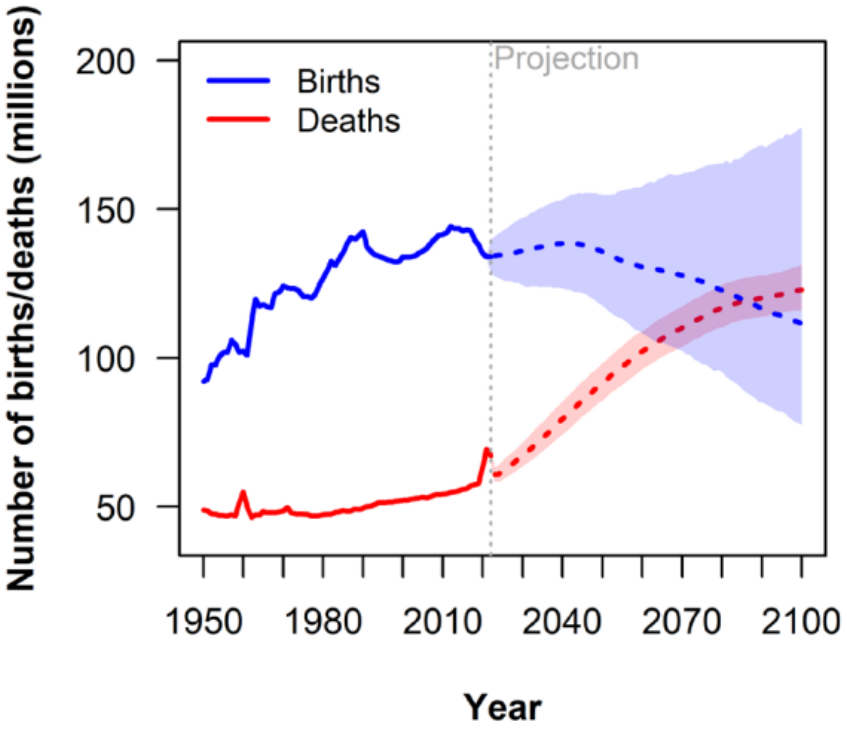


United Nations,
Department of Economic and Social Affairs

Global death rate and comparison with Sub-Saharan Africa

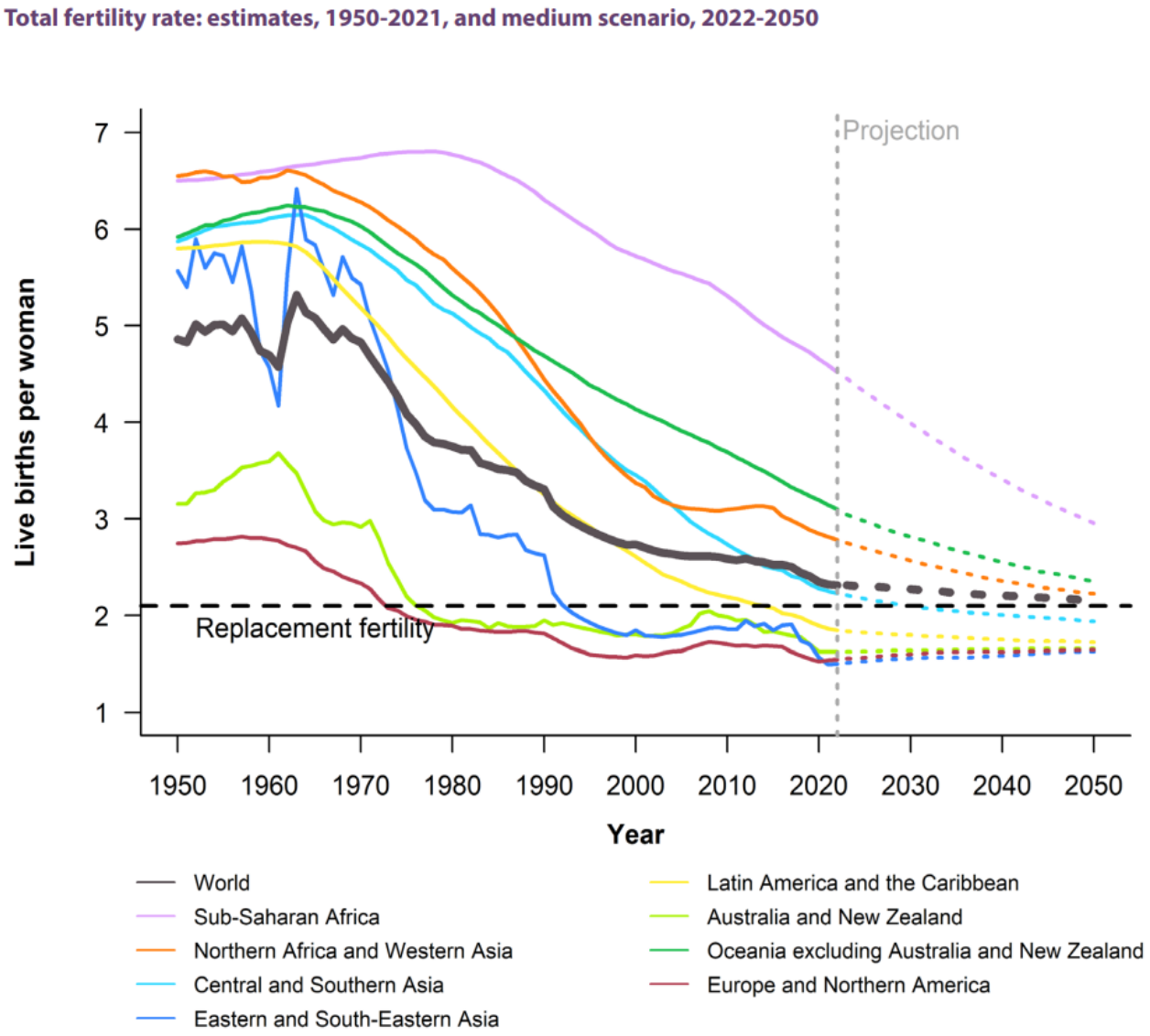


Source: UN DESA, 2022.

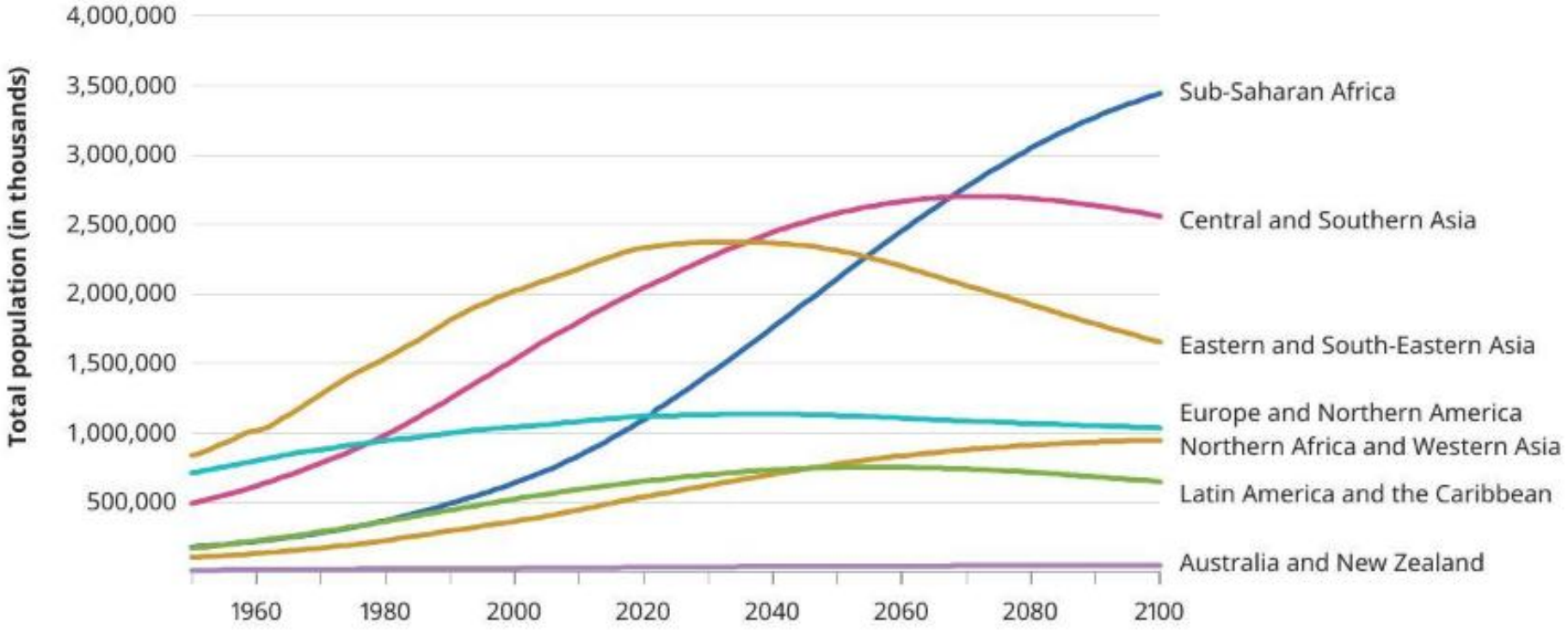


United Nations,
Department of Economic and Social Affairs

Global fertility rates of some countries and regions in the world



Growth in total population in different regions of the world



Source: UNFPA Technical Division, 2023.

United Nations Population

Energy Sources

- Modern society requires large quantities of energy that are generated from the Earth's natural resources.
- **Primary Energy Resources**: The fossil fuels (oil, gas and coal), nuclear energy, falling water, geothermal, wind and solar energy.
- Secondary Energy Resources: Those sources which are derived from primary resources such as electricity, fuels from coal, (synthetic natural gas and synthetic gasoline).

Energy Sources

- Energy sources are divided into two groups:

- **Renewable**

  (an energy source that can be easily replenished)

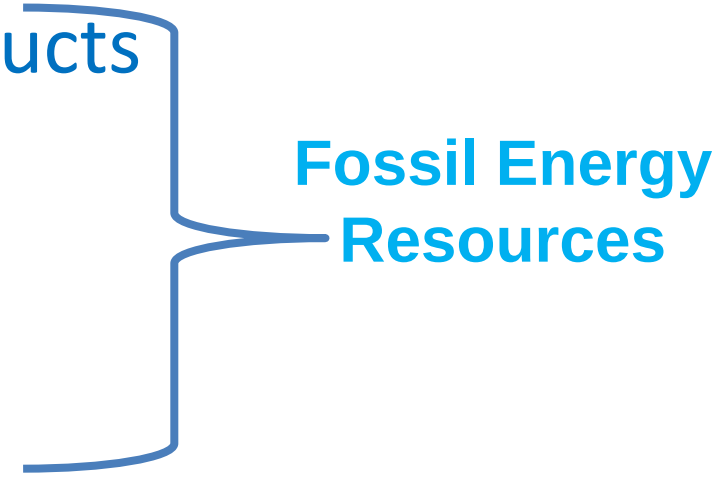
- **Nonrenewable**

  (an energy source that cannot be easily replenished)

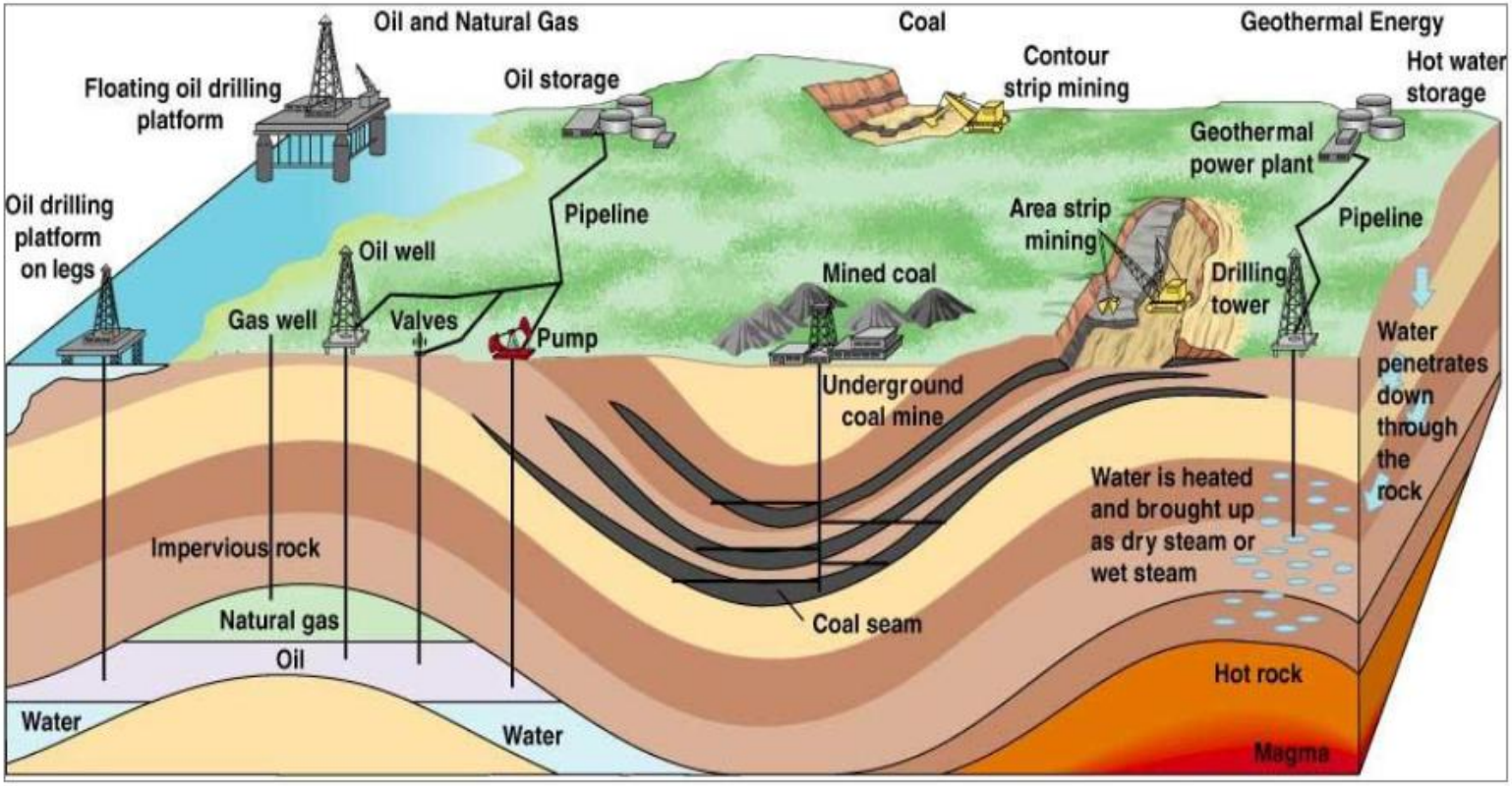
Nonrenewable Energy Sources

Nonrenewable energy sources are:

- Petroleum & petroleum products
- Natural Gas
- Hydrocarbon gas liquids
- Coal
- Nuclear energy



Energy resources in Earth's crust

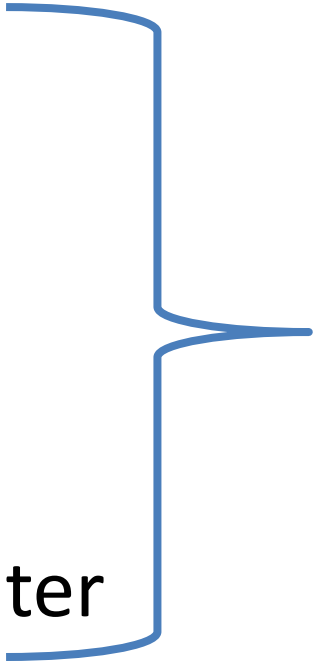


- Energy resources removed from the earth's crust include: ***oil, natural gas, coal, and uranium and thorium***
- The other raw materials are dependent on your geographic position ; ***sun, wind, geothermal and hydro***

Renewable Energy

There are five main renewable energy sources:

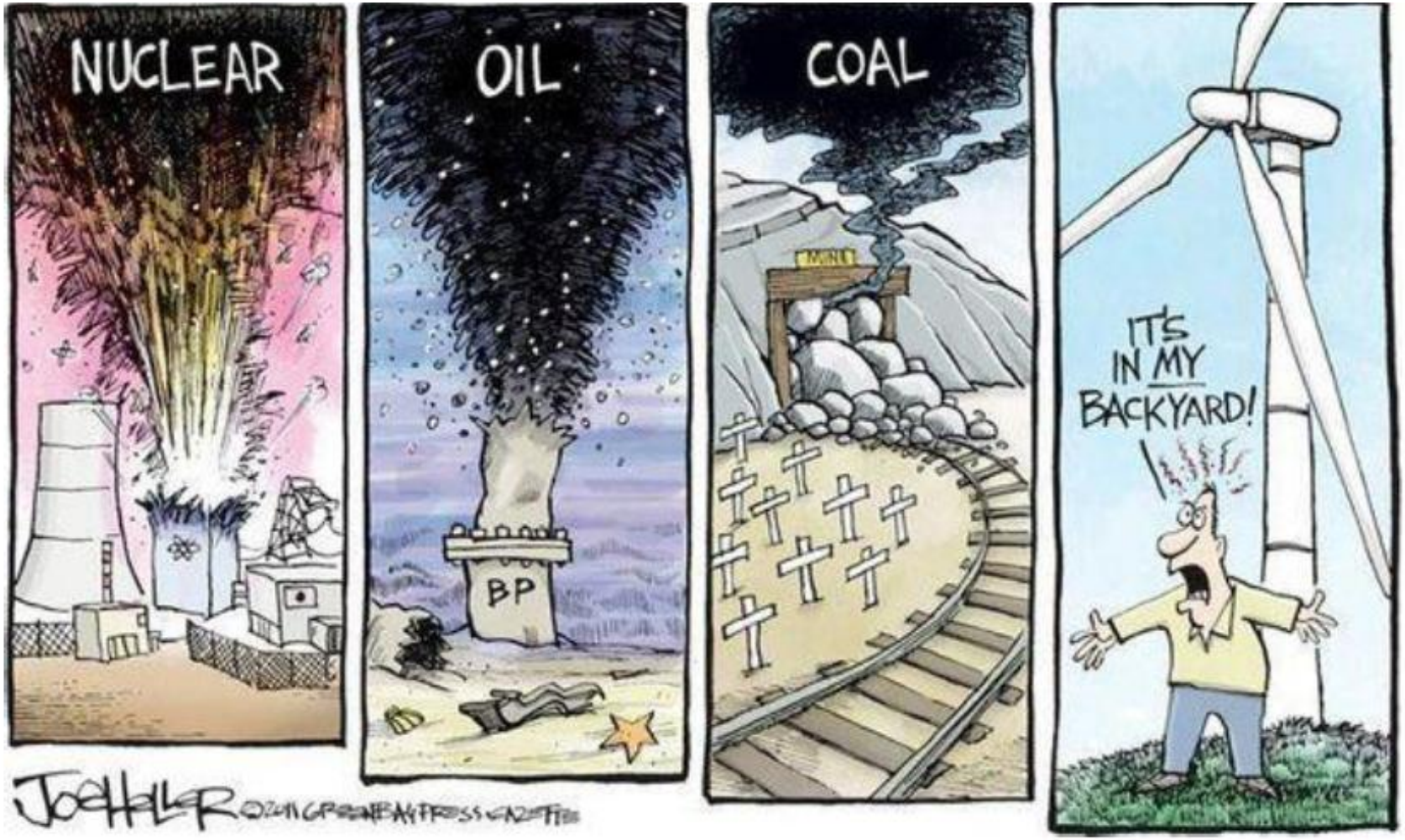
- Solar energy from the sun
- Wind energy
- Wave Energy
- Biomass from plants
- Hydropower from flowing water
- Geothermal energy from heat **inside** the earth



Renewable Energy

- They are dependent of the geographic position of the location and cannot be transported from one place to another for to use; so that they are not strategic
- They can be reused and they are carbon free raw materials so that they are environmentally safe
- They can generate electricity but their supply is changeable due to seasons, or other reasons so they are not sustainable for large electricity consumers except hydro and geothermal
- Still there are technology concerns on these energy resources
- They are not efficient for to supply the whole consumption need of a country or even a factory
- Their efficiency is related to weather conditions
- Still countries invest into their renewable energy capacity for reducing their risk in supply security in fossil fuels and nuclear
- Sun, wind, hydro, geothermal are considered to be renewable energy resources
- They are abundant but not infinite, however they are environment friendly and carbon free

Arguments against the energy resources



Energy Units

- Barrels or gallons for liquid petroleum fuels (such as gasoline, diesel fuel, and jet fuel) and biofuels (ethanol and biodiesel)
- Cubic feet for natural gas (0.02832 cubic meter or 28.32 lt)
- Cubic meter for natural gas
- Tons for coal
- Kilowatthours for electricity ($1m^3$= 10.64 kWh)

Units for comparing energy

- British thermal units (Btu),
- Barrels of oil equivalent (Boe),
- Metric tons of oil equivalent,
- Metric tons of coal equivalent,
- Terajoules (277 777.777 777 78 kWh).

- **A *British thermal unit* (Btu) is a measure of the heat content of fuels or energy sources. It is the quantity of heat required to raise the temperature of one pound of liquid water by 1 degree Fahrenheit at the temperature that water has its greatest density (approximately 39 degrees Fahrenheit).**

Btu content of common energy units

- 1 barrel (42 gallons) of crude oil = 159 lt
- 1 barrel (42 gallons) of crude oil = 5,717,000 Btu
- 1 gallon of gasoline = 120,476 Btu
- 1 gallon of diesel fuel = 137,452 Btu
- 1 cubic feet of natural gas = 1,037 Btu
- 1 ton (2,000 pounds) of coal = 19,146,000 Btu
- 1 kilowatthour of electricity = 3,412 Btu


THANK YOU